Automation Tech-Tips

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Title: Turbidity and the Measuring Principle of the 4150 Turbidimeter

In today's world of increasing government regulations it is becoming more important to deploy reliable analytical instruments which are compliant to certain laws and directives, such as the Signet 4150 Turbidimeter. This is especially true for the drinking water, wastewater as well as food and beverage industries, where hygiene and health protection play an essential role. But what exactly is Turbidity? What are the government regulations. How does the Signet 4150 Turbidimeter measure the turbidity of the water?

Definition of Turbidity

Turbidity is the cloudiness of a liquid caused by the presence of individual particles (suspended solids and colloidal matter) that are generally invisible to the naked eye, similar to smoke in air. Suspended materials include dead and living organic material such as plankton, algae, microbes as well as particles from mineral origin such as clay, silt, and sand, and other substances. Turbidity indicates the clarity of water and its measurement is therefore a key test of water quality. These particles can also affect the color of the water. Raw water filtered to produce drinking water must have suspended materials removed during the treatment process.

0.02 NTU and 1000 NTU Turbidity Cuvette

The most common measurement units for turbidity are NTU (Nephelometric Turbidity Units), FTU (Formazin Turbidity Units), or for ISO standards, FNU (Formazin Nephelometric Units).

Government Regulations

Most countries' governments have established certain standards for the allowable turbidity rating of drinking water and the method for regulatory water quality monitoring.

In the United States, where the white light method (according to U.S. EPA 180.1) is accepted to measure turbidity, the turbidity cannot be higher than 1.0 NTU at the plant outlet after conventional or direct filtration methods, and all samples for turbidity must be less than or equal to 0.3 NTU for at least 95 percent of the samples in any month. Other systems must follow state limits, where turbidity cannot exceed 5 NTU at any time.
In other countries, measuring methods and limits for Nephelometric turbidity units may be different from the United States. For example, the infrared light Turbidimeter versions according to ISO 7027 are most commonly used in Europe.

The Signet 4150 Turbidimeter, with its infrared light and white light versions, complies to domestic and international government regulations for monitoring water quality. To ensure that you choose the correct version for your application, we recommend checking with your regulatory agency for local requirements.

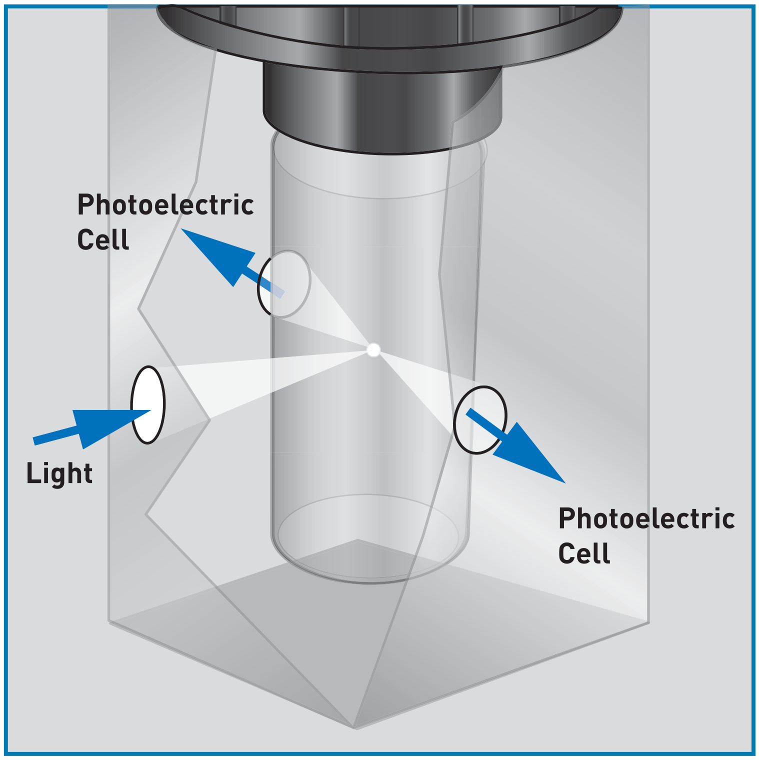
Theory of Operation

The Signet 4150 Turbidimeter measures turbidity via the standardized 90 degrees scattered light method [1]. The method is based upon a comparison of the intensity of light scattered by the sample under defined conditions with the intensity of light scattered by a standard reference suspension [2].

In the Signet 4150 Turbidimeter, turbidity is measured by directing a light beam to a measuring sample cuvette. This light beam is scattered by the suspended particles as they flow through the medium sample in the small volume glass cuvette, and detected by two photoelectric cells that are located at 90 degrees angle to the light source. The amount of scattered light determines the turbidity of the medium. The higher the intensity of the scattered light, the more particles in the medium, and the higher the turbidity. The measurement optics of the 4150 Turbidimeter are not directly in touch with the medium sample and therefore the instrument requires only little maintenance. The units of turbidity are displayed in Nephelometric Turbidity Units (NTU), or alternatively in Formazin Nephelometric Units (FNU).

[1] U.S. EPA 180.1 for white light versions and ISO 7027 / EN 27027 for infrared light versions

[2] U.S. EPA: Method 180.1 Determination of Turbidity by Nephelometry, 1993



90° scattered light method

Ordering Information

Mfr. Part No.	Code	Description
3-4150-1	159 001 596	Turbidimeter, White Light, 0 to 1000 NTU/FNU, without ultrasonic cleaning option
3-4150-2	159 001 597	Turbidimeter, Infrared Light, 0 to 1000 NTU/FNU without ultrasonic cleaning option
3-4150-3	159 001 598	Turbidimeter, White Light, 0 to 100 NTU/FNU, with ultrasonic cleaning option
3-4150-4	159 001 599	Turbidimeter, Infrared Light, 0 to 100 NTU/FNU with ultrasonic cleaning option
3-4150-5	159 001 600	Turbidimeter, White Light, 0 to 1000 NTU/FNU with ultrasonic cleaning option
3-4150-6	159 001 601	Turbidimeter, Infrared, 0 to 1000 NTU/FNU with ultrasonic cleaning option

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